

MikiLab v14 - Ecosystem Document

Complete documentation of the Cyber-Industrial ecosystem (64 modules). Language: English.

Architecture & Philosophy

MikiLab is a hands-free digital lab for bakers: 100% offline-first, continuous voice synthesis without keywords, Cyber-Industrial teal/dark style. Guidance is handled exclusively by the dynamic avatars of the Cyber Trio (no static text). No HACCP/allergens.

1. The 8 Macro-Areas

Home

Lean showcase with navigation text, interactive Cyber-Bakery Trio, Miki's Board and the Talk with Miki assistant.

Chef/Lab Mode

Operational heart: machine park with global timers, hands-free Voice Core, Commander Bridge, Smart Planner, Thermal Guard, Team OS and 50+ tools.

My Recipes

Tested recipes with grams and baker's percent, method, timings and cost; create and manage your own recipes.

Science & Guides (Learn)

Step-by-step path, Bread Sommelier, technical fermentation cards, Cyber-Bakery Trio with mini-guides.

Community

Persistent Pro social feed, channels to follow, exchange among bakers.

Manual & B2B Operations

Machine compendium + hands-free voice commands + recommended hardware + index of modules 1-64.

Commander Bridge

AI reporting, team radar, predictive maintenance and Multi-Chief Executive Architecture (module 64).

Talk with Miki

First-person conversational AI assistant (voice + text), with co-pilot Miki/Mohamed/Bake Mix, live intercom and continuous listening.

2. Cyber Trio - Operational Cards

Miki (The Commander)

Lean founder, shaved head, earring, a real tattoo on the left arm, MikiLab t-shirt. Conversational AI assistant: answers in first person about how the software helps bakers and about the whole lab.

Mohamed (The Right Hand)

Young Persian/Iranian man in MikiLab uniform. Handles practical operations (cleaning, trolleys, loading) with the 'Lab Live View' (micro-animations of operations).

Bake Mix (Robot Assistant)

Futuristic baker robot in teal/amber. Handles technology, AI and voice commands; leads the Interactive Training of hands-free commands.

Shared Trio features:

- 1 Intercom over industrial Bluetooth headsets

- 2 Audio Auto-Duck (lowers the radio during announcements, then smooth return to 100%)
- 3 Smart Context Memory (per-session conversation history)
- 4 Dynamic handoff animation when switching co-pilot
- 5 Empathetic step-by-step support for staff
- 6 Clean speech synthesis (TTS) with synchronized haptics

3. Commander Bridge - Commander Voice

- 1 Automatic and voice logging of batches, recipes, ovens and calendars
- 2 Multi-Chief: profile of the on-duty commander (name, tone, priorities)
- 3 Automatically generated shift-handover briefing (voice + visual)
- 4 Tactical Bottleneck AI and Neural Load Radar with glow on the critical parameter
- 5 IoT Predictive Maintenance and Downtime Root-Cause Logging
- 6 End-of-Day Report (backend storage, historical chart, automatic evening save)

4. Map of the 64 Operational Modules

#	Module	#	Module
1	Predictive Flour Hydration AI	33	Multi-Feeder Guard
2	Dynamic Oven Thermal Planner	34	Rheon PMU Recall
3	Thermal Timeline Red/Amber	35	Thermal Friction Index
4	Live Energy & kWh Counter	36	Hydraulic Press Pressure & Thickness
5	Tactical Bottleneck AI	37	Multi-Deck & Rotary Oven Airflow Hydro-Balance
6	Neural Load Radar	38	Amperometric Torque Control
7	Cyber-Wave Audio Visualizer	39	Volumetric Divider & Piston Wear
8	IoT Predictive Maintenance	40	Dynamic Climate Control Proofer
9	Automated Yeast Maturity Tracker	41	Predictive Loading Bay & Route Dispatch
10	Cold Chain Auto-Recovery	42	Dynamic Rack & Trolley Buffer Tracking
11	Energy Tariff Auto-Shift	43	Synchronized Multi-Oven Unloading Wave
12	Voice Batch Traceability	44	Real-Time Ingredient Buffer Replenishment
13	Real-Time Yield & Waste Analytics	45	Executive Emergency Re-Routing AI
14	Multi-Zone Audio Routing	46	Dynamic Tray, Mold & Oven Tool Inventory
15	Dynamic Delivery Express Sync	47	Steam Recovery & Recirculation
16	Ghost Shift Auto-Pilot	48	Hands-Free Quick Maintenance Guide
17	Virtual Baker Coach	49	Downtime Root-Cause Logging
18	Digital Silo Telemetry	50	Multi-Deck Load Balancing AI
19	Equipment Stress Index	51	Startup Splash Screen (Glitch-Neon)
20	Voice Shift Handover Notes	52	Miki's Board
21	Offline Spatial Tour	53	Cyber-Industrial Kit (Space-Tech font)
22	Task Decomposition	54	Synchronized Success Sound
23	Offline Voice Glossary	55	Dynamic Focus Animation
24	Micro-Validation	56	Miki's Handshake (haptic)
25	Apprentice Progress Index	57	Mohamed's Lab Live View
26	Four-Hand Co-Weighing	58	Bake Mix - Interactive Training
27	SOS Colleague	59	Audio Snapshot
28	Voice Kudos	60	Smart Scale Sync
29	Shaping Metronome	61	Stress Index
30	Cross-Shift Handover Log	62	Recipe Scaling
31	Rheon Dough/Filling Ratio	63	Squad Check-In
32	Shutter Speed Sync	64	Multi-Chief Executive Architecture

5. Active Lab Tools (Cards)

6. Equipment & B2B Hardware Guide

Recommended IoT sensors & hardware

Product	Use	Where to get it
Milesight EM300-TH	Temperature + humidity LoRaWAN probe for cells and environment.	milesight.com
Efento (NB-IoT / BLE)	Temperature data logger for the cold chain (freezer/fridge).	getefento.com
Probes PT100 / DS18B20	Contact probes for dough core and oven (via gateway/ESP).	rs-online.com
Relay Shelly / Sonoff	WiFi relays to drive ovens, lights and cell heating.	shelly.com
Jabra wireless headset	Noise-cancelling headset for the hands-free Voice Core.	jabra.com

Integrable machines

Machine	Function & Modules	Brands / Sources
Rheon lines (Encrusting/Extruder)	Dough/filling dosing and continuous forming. Modules 31-35.	Rheon
Hydraulic presses & molders	Pressure/thickness calibration with elasticity control. Module 36.	Bakery equipment vendors
Multi-deck & rotary ovens	Air/steam balance, steam recovery, load-balancing AI. Modules 37/47/50.	Wachtel, MIWE, Zucchelli
Spiral mixers & dividers	Amperometric torque, stress index, piston wear. Modules 38/19/39.	Pro mixer vendors
Proofing & freezing cells	Dynamic climate and cold-chain recovery. Modules 40/10.	Cell & blast-chiller vendors
IoT scales (Smart Scale)	Guided weighing and four-hand co-weighing, smart-scale sync. Modules 26/60.	Connected industrial scales
Industrial flour silos	Level and flour-flow telemetry. Module 18.	Silo & flour-transport vendors
Trays, molds, peels & tools	Cycle, cleaning and sanitizing tracking. Module 46.	Bakery tooling vendors

7. Real website screenshots (Illustrated manual)



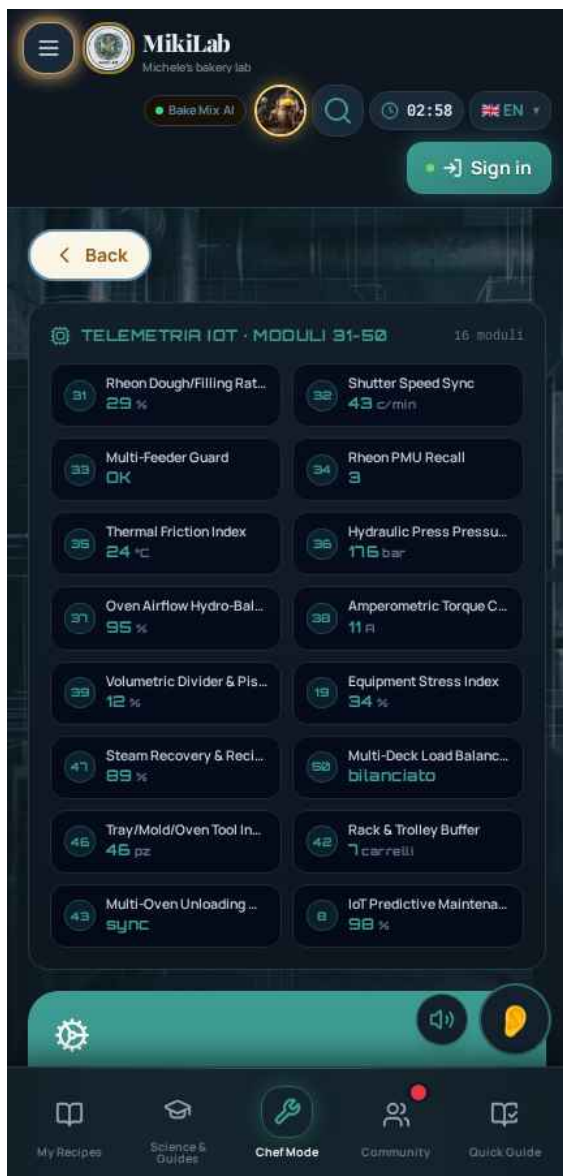
Home - Navigation, Cyber-Bakery Trio and Miki's Board



Talk with Miki - AI assistant, co-pilot, live intercom, continuous listening



Commander Bridge - AI reporting, glow on the critical parameter, Multi-Chief (module 64)



Machine Park - IoT telemetry, modules 31-50 with live parameters



Manual & B2B Operations - Lab usage and voice commands